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Inventor(s)	Susca; Ryan Prescott et al.

FUEL PUMP SYSTEM WITH BUILT-IN THERMAL BYPASS

Abstract

A fuel delivery system for a gas turbine engine includes a fuel source, and a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine. A fuel oil cooler is positioned fluidly between the main fuel pump and the combustor assembly configured to cool oil utilized by the fuel delivery system, and a fuel oil cooler bypass pathway is configured to selectably direct the first flow of fuel to bypass the fuel oil cooler.

Inventors: **Susca; Ryan Prescott (Windsor, CT), Shook; Ryan (Loves Park, IL)**

Applicant: **Hamilton Sundstrand Corporation** (Charlotte, NC)

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Background/Summary

BACKGROUND

[0001] Exemplary embodiments pertain to the art of gas turbine engines, and more particularly to fuel delivery systems of gas turbine engines.

[0002] Fuel delivery systems for gas turbine engines utilize a variety of pumps, often driven by a

gearbox that extracts power from operation of the gas turbine engine to drive the pumps. The pumps include main pumps and an actuation pump. Modern aircraft require more efficient pumping systems and higher running fuel temperatures to improver overall thrust specific fuel consumption (TSFC), which is one measure of operational efficiency of the gas turbine engine and the aircraft.
[0003] There is a desire to reduce fuel system gearbox horsepower extraction, but in single engine aircraft it is difficult to move away from inefficient centrifugal main pumps because of their high reliability. These aircraft struggle with satisfying fuel system thermal management requirements at low power conditions when centrifugal pumps are at their most inefficient.
BRIEF DESCRIPTION
[0004] In one exemplary embodiment, a fuel delivery system for a gas turbine engine includes a fuel source, and a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine. A fuel oil cooler is positioned fluidly between the main fuel pump and the combustor assembly configured to cool oil utilized by the fuel delivery system, and a fuel oil cooler bypass pathway is configured to selectably direct the first flow of fuel to bypass the fuel oil cooler.
[0005] Additionally or alternatively, in this or other embodiments a fuel oil cooler bypass valve is positioned along the fuel oil cooler bypass pathway to control the first flow of fuel along the fuel oil cooler bypass pathway.
[0006] Additionally or alternatively, in this or other embodiments the fuel oil cooler bypass valve is one of actively controlled or thermostatically controlled.
[0007] Additionally or alternatively, in this or other embodiments a main fuel filter is located upstream of the main fuel pump, and an actuation pump is positioned upstream of the main fuel filter and is configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.
[0008] Additionally or alternatively, in this or other embodiments an augmentor fuel pump is positioned upstream of the main fuel filter and is configured to direct a third flow of fuel to an augmentor of the gas turbine engine.
[0009] Additionally or alternatively, in this or other embodiments a boost pump is positioned upstream of the main fuel filter and the actuation pump.
[0010] Additionally or alternatively, in this or other embodiments the second flow of fuel is urged from the one or more actuation devices to the main fuel filter.
[0011] In another exemplary embodiment, a gas turbine engine and fuel delivery system includes a gas turbine engine, including a turbine, and a combustor where a first flow of fuel is combusted to drive the turbine via a flow of combustion products. A fuel delivery system is operably connected to the gas turbine engine and includes a fuel source, and a main fuel pump configured to deliver a first flow of fuel to the combustor. A fuel oil cooler is positioned fluidly between the main fuel pump and the combustor configured to cool oil utilized by the fuel delivery system, and a fuel oil cooler bypass pathway is configured to selectably direct the first flow of fuel to bypass the fuel oil cooler.
[0012] Additionally or alternatively, in this or other embodiments a fuel oil cooler bypass valve is positioned along the fuel oil cooler bypass pathway to control the first flow of fuel along the fuel oil cooler bypass pathway.
[0013] Additionally or alternatively, in this or other embodiments the fuel oil cooler bypass valve is one of actively controlled or thermostatically controlled.
[0014] Additionally or alternatively, in this or other embodiments a main fuel filter is positioned upstream of the main fuel pump, and an actuation pump is positioned upstream of the main fuel filter and is configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.
[0015] Additionally or alternatively, in this or other embodiments an augmentor fuel pump is positioned upstream of the main fuel filter and configured to direct a third flow of fuel to an

augmentor of the gas turbine engine.
[0016] Additionally or alternatively, in this or other embodiments a boost pump is positioned upstream of the main fuel filter and the actuation pump.
[0017] Additionally or alternatively, in this or other embodiments the second flow of fuel is urged from the one or more actuation devices to the main fuel filter.
[0018] In another exemplary embodiment, a method of operating a fuel delivery system of a gas turbine engine includes urging a first flow of fuel from a fuel source and through a main fuel pump, directing the first flow of fuel from the main fuel pump toward a combustor assembly of the gas turbine engine, directing the first flow of fuel through a fuel oil cooler disposed fluidly between the main fuel pump and the combustor assembly, and selectably flowing the first flow of fuel through a fuel oil cooler bypass pathway to bypass the fuel oil cooler.
[0019] Additionally or alternatively, in this or other embodiments a fuel oil cooler bypass valve positioned along the fuel oil cooler bypass pathway is operated to control the first flow of fuel along the fuel oil cooler bypass pathway.
[0020] Additionally or alternatively, in this or other embodiments the first flow of fuel is flowed through a main fuel filter positioned upstream of the main fuel pump, and a second flow of fuel is flowed through an actuation pump positioned upstream of the main fuel filter. The actuation pump is configured to deliver the second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.
[0021] Additionally or alternatively, in this or other embodiments a third flow of fuel is directed through an augmentor fuel pump positioned upstream of the main fuel filter toward an augmentor of the gas turbine engine.
[0022] Additionally or alternatively, in this or other embodiments at least the first flow of fuel is flowed through a boost pump positioned upstream of the main fuel filter and the actuation pump.
[0023] Additionally or alternatively, in this or other embodiments the second flow of fuel is urged from the one or more actuation devices to the main fuel filter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
[0025] FIG. **1** is a schematic illustration of an embodiment of a gas turbine engine and fuel delivery system through which fuel is delivered to power the gas turbine engine;
[0026] FIG. **2** is a schematic illustration of an embodiment of a fuel delivery system of a gas turbine engine; and
[0027] FIG. **3** is a schematic illustration of another embodiment of a fuel delivery system of a gas turbine engine.
DETAILED DESCRIPTION
[0028] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
[0029] An aircraft gas turbine engine system is indicated generally at **10** in FIG. **1**. Gas turbine engine system **10** includes a compressor portion **12** operatively coupled to a turbine portion **14** through a shaft **16**. A combustor assembly **18** is fluidically connected between the compressor portion **12** and turbine portion **14**. A fuel delivery system **20** fluidically connects the combustor assembly **14** with a source of fuel **24**. Fuel delivery system **20** may receive fuel directly from source of fuel **24** or, through a compressor stage (not shown) that creates an input pressure for the fuel. In addition to the combustor assembly **18**, the fuel delivery system **20** is operably connected to one or more secondary components, such as an augmentor **26** to deliver a flow of fuel thereto. In

some embodiments, a gearbox **28** is operably connected to the gas turbine engine **10** via, for example, the shaft **16** to extract power from the gas turbine engine **10**. The fuel delivery system **20** is operably coupled to the gearbox to power components of the fuel delivery system **20**.

[0030] Referring now to FIG. **2**, an embodiment of a fuel delivery system **20** will now be described. In the embodiment of FIG. **2**, the fuel delivery system **20** is a three-pump fuel delivery system **20**, with independently operating main fuel pump **30**, augmentor fuel pump **32** and actuation fuel pump **34**. Initial a flow of fuel from the fuel source **24** is directed through a boost pump **36** at which the pressure of the flow of fuel is increased. From the boost pump **36**, the flow of fuel is directed toward the main fuel pump **30** and the augmentor fuel pump **32** along a fuel pathway **38**. A main portion **40** of the flow of fuel flows through a main fuel filter **42** and then through the main fuel pump **30** toward the combustor assembly **18**. The flow of the main portion **40** is controlled by one or more main control valves **50**. An actuation portion **44** of the flow of fuel, once downstream of the main fuel pump **30**, is directed toward the actuation fuel pump **34**. The actuation portion **44** is urged toward one or more actuation devices **46** of the fuel delivery system **20** to power the actuation devices **46** at, for example, the combustor assembly **18** and/or the augmentor **26**. From the actuation devices **46**, the actuation portion **44** is recirculated to the main fuel filter **42**, or alternatively to the augmentor fuel pump **32**. The flow of the actuation portion **44** is controlled by one or more actuation control valves **48**.

[0031] Before the main portion **40** reaches the main fuel filter **42**, a part of the main portion **40**, designated as an augmentor portion **52** of the flow of fuel, is directed toward the augmentor fuel pump **32** and from the augmentor fuel pump **32** toward the augmentor **26**. The flow of the augmentor portion **52** is controlled by one or more augmentor control valves **54**. Excess fuel of the augmentor portion **52** may be returned from the augmentor **26** to, for example, the augmentor fuel pump **32** or the boost pump **36**.

[0032] Downstream of the main fuel pump **30** between the main fuel pump **30** and the combustor assembly **18**, a fuel oil cooler (FOC) **56** is positioned to cool lubricant utilized by the gearbox **28** and/or pumps **30** and **32** via the flow of the main portion **40** through the FOC **56**. For optimal gearbox **28** performance, it is desired that a temperature of the oil is maintained within a preselected range. In some operating conditions of the fuel delivery system **20**, however, the FOC **56** may operate to cool the oil such that the temperature of the oil is below the preselected range. To counteract this potential overcooling of the oil, the fuel delivery system includes an FOC bypass pathway **58** that selectably directs the main portion **40** around the FOC **56** so that the main portion **40** does not act to cool the oil at the FOC **56**. Flow to the FOC bypass pathway **58** is controlled by operation of an FOC bypass valve **60** located along the FOC bypass pathway **58**. The FOC bypass valve **60** may be operated by, for example, an electrohydraulic servo valve **62** and/or a linear variable differential transformer **64** operably connected to the FOC bypass valve **60**. In some embodiments, the FOC bypass valve **60** is actively controlled via connected to a temperature sensor **66** that measures a temperature of the oil at the FOC **56**, and when the oil temperature falls below the preselected range, the FOC bypass valve **60** is opened, thus directing the main portion **40** along the FOC bypass pathway **58**. When the oil temperature returns to within the preselected range, the FOC bypass valve **60** may be signaled to close, thus directing the main portion **40** through the FOC **56**.

[0033] Another embodiment is illustrated in FIG. **3**, where the fuel delivery system **20** is a two-pump system having an actuation fuel pump **34** and a combined main fuel pump and augmentor fuel pump **68**.

[0034] The FOC **56** is located downstream of the combined fuel pump **68** to cool the oil utilized by the gearbox **28** and/or the combined fuel pump **68**. Flow to the FOC bypass pathway **58** is controlled by operation of the FOC bypass valve **60** located along the FOC bypass pathway **58**. The FOC bypass valve **60** may be operated by, for example, an electrohydraulic servo valve **62** and/or a linear variable differential transformer **64** operably connected to the FOC bypass valve **60**. In some embodiments, the FOC bypass valve **60** is actively controlled via connected to a temperature sensor **66** that measures a temperature of the oil at the FOC **56**, and when the oil temperature falls below the preselected range, the FOC bypass valve **60** is opened, thus directing the main portion **40** along the FOC bypass pathway **58**. When the oil temperature returns to within the preselected range, the FOC bypass valve **60** may be signaled to close, thus directing the main portion **40** through the FOC **56**.

[0035] Utilization of the FOC bypass pathway **58** and the FOC bypass valve **60** allows for improved control of the temperature of the oil, and prevents overcooling of the oil by the FOC **56**. This improved control of the oil temperature improves pump and gearbox performance.

[0036] The term "about" is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

[0037] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

[0038] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A fuel delivery system for a gas turbine engine, comprising: a fuel source; a main fuel pump configured to deliver a first flow of fuel to a combustor assembly of the gas turbine engine; a fuel oil cooler disposed fluidly between the main fuel pump and the combustor assembly configured to cool oil utilized by the fuel delivery system; and a fuel oil cooler bypass pathway configured to selectably direct the first flow of fuel to bypass the fuel oil cooler.

2. The fuel delivery system of claim 1, further comprising a fuel oil cooler bypass valve disposed along the fuel oil cooler bypass pathway to control the first flow of fuel along the fuel oil cooler bypass pathway.

3. The fuel delivery system of claim 1, wherein the fuel oil cooler bypass valve is one of actively controlled or thermostatically controlled.

4. The fuel delivery system of claim 1, further comprising: a main fuel filter disposed upstream of the main fuel pump; and an actuation pump disposed upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.

5. The fuel delivery system of claim 4, further comprising an augmentor fuel pump disposed upstream of the main fuel filter and configured to direct a third flow of fuel to an augmentor of the gas turbine engine.

6. The fuel delivery system of claim 4, further comprising a boost pump positioned upstream of the main fuel filter and the actuation pump.

7. The fuel delivery system of claim 4, wherein the second flow of fuel is urged from the one or more actuation devices to the main fuel filter.

8. A gas turbine engine and fuel delivery system, comprising: a gas turbine engine, including: a turbine; and a combustor where a first flow of fuel is combusted to drive the turbine via a flow of combustion products; and a fuel delivery system operably connected to the gas turbine engine, including: a fuel source; a main fuel pump configured to deliver a first flow of fuel to the combustor; a fuel oil cooler disposed fluidly between the main fuel pump and the combustor configured to cool oil utilized by the fuel delivery system; and a fuel oil cooler bypass pathway configured to selectably direct the first flow of fuel to bypass the fuel oil cooler.

9. The gas turbine engine and fuel delivery system of claim 8, further comprising a fuel oil cooler bypass valve disposed along the fuel oil cooler bypass pathway to control the first flow of fuel along the fuel oil cooler bypass pathway.

10. The gas turbine engine and fuel delivery system of claim 8, wherein the fuel oil cooler bypass valve is one of actively controlled or thermostatically controlled.

11. The gas turbine engine and fuel delivery system of claim 8, further comprising: a main fuel filter disposed upstream of the main fuel pump; and an actuation pump disposed upstream of the main fuel filter and configured to deliver a second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.

12. The gas turbine engine and fuel delivery system of claim 11, further comprising an augmentor fuel pump disposed upstream of the main fuel filter and configured to direct a third flow of fuel to an augmentor of the gas turbine engine.

13. The gas turbine engine and fuel delivery system of claim 11, further comprising a boost pump positioned upstream of the main fuel filter and the actuation pump.

14. The gas turbine engine and fuel delivery system of claim 11, wherein the second flow of fuel is urged from the one or more actuation devices to the main fuel filter.

15. A method of operating a fuel delivery system of a gas turbine engine, comprising: urging a first flow of fuel from a fuel source and through a main fuel pump; directing the first flow of fuel from the main fuel pump toward a combustor assembly of the gas turbine engine; directing the first flow of fuel through a fuel oil cooler disposed fluidly between the main fuel pump and the combustor assembly; and selectably flowing the first flow of fuel through a fuel oil cooler bypass pathway to bypass the fuel oil cooler.

16. The method of claim 15, further comprising operating a fuel oil cooler bypass valve disposed along the fuel oil cooler bypass pathway to control the first flow of fuel along the fuel oil cooler bypass pathway.

17. The method of claim 15, further comprising: flowing the first flow of fuel through a main fuel filter disposed upstream of the main fuel pump; and flowing a second flow of fuel through an actuation pump disposed upstream of the main fuel filter, the actuation pump configured to deliver the second flow of fuel from the fuel source to one or more actuation devices of the gas turbine engine.

18. The method of claim 17, further comprising directing a third flow of fuel through an augmentor fuel pump disposed upstream of the main fuel filter toward an augmentor of the gas turbine engine.

19. The method of claim 17, further comprising flowing at least the first flow of fuel through a boost pump positioned upstream of the main fuel filter and the actuation pump.

20. The method of claim 17, further comprising urging the second flow of fuel from the one or more actuation devices to the main fuel filter.